RAJVEL R

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SUMMARY

AI Engineer with 2+ years of experience building scalable backend systems and, most recently, production-grade agentic AI platforms. Currently architecting multi-agent systems at Blackstraw.ai — orchestrator-managed sub-agents, A2A protocol communication, SPIFFE/SPIRE-based agent identity, OAuth/OIDC security, and MCP server deployments on GCP and Kubernetes. Combines strong distributed-systems fundamentals (FastAPI, PostgreSQL, Redis, Docker, Azure) with hands-on GenAI/LLM engineering, RAG pipelines, and agent evaluation.

EXPERIENCE

SDE-III — AI/Agentic Systems | Blackstraw.ai **Jun 2026 – Present**

Multi-Agent Systems · A2A Protocol · SPIFFE/SPIRE · OAuth 2.0/OIDC · MCP · GCP (Cloud Run, GKE) · Agent Evaluation

- Built multi-agent systems with an orchestrator managing specialised sub-agents, writing clear system prompts and instructions for each agent using proven prompting techniques.
- Set up agent-to-agent communication using the A2A protocol, with SPIFFE/SPIRE-based identity for agents that need a unique, verifiable identity.
- Added OAuth 2.0 and OIDC based login and permissions for the agent platform, so both users and individual agents are securely authenticated.
- Built memory into the agent systems so agents keep context across sessions instead of starting fresh each time.
- Set up agent evaluation combining GCP online monitoring, offline evaluation with golden datasets, and LLM-as-a-judge scoring to verify prompts and validate agent flows before release.
- Built a dynamic orchestrator that picks the right workflow — sequential, parallel, loop, or custom — at runtime depending on the task.
- Built a standalone Agent Development Kit (ADK) package, similar to Google's ADK, with built-in logging, tracing, and observability for agent workflows.
- Built MCP servers and tools, deployed to Cloud Run and on-prem Kubernetes, so agents could call external/third-party APIs as tools; connected securely to GCP-hosted agents using Workload Identity Federation (WIF).
- Tracked and optimised model usage costs across agents, and cached conversation memory in Redis to cut agent response latency and avoid slow reloads of session history.
- Added guardrails and content-safety checks on both incoming prompts and outgoing responses, with GCP Cloud Armor for prompt-injection defense, plus distributed tracing and observability via Langfuse.
- Built the agent-facing frontend using CopilotKit's AG-UI protocol, enabling real-time streaming of agent responses and actions into the UI.

Software Engineer – Backend | Populus Empowerment Network **Jan 2024 – Jun 2026**

Python · FastAPI · SQLAlchemy · PostgreSQL · Redis · Celery · Docker · Azure · Loki · Grafana

- Reduced P95 API latency by 40% with Redis caching and query optimisation on high-traffic endpoints.
- Solely owned migration of 65M+ MSSQL records to PostgreSQL, completing it 80% faster than estimated.
- Architected a unified JWT + RBAC auth layer for a multi-tenant platform with zero production regressions.
- Built and owned a Loki + Grafana observability stack across 6+ microservices as primary on-call engineer.
- Designed and deployed a RAG pipeline using LangChain, pgvector, and sentence-transformers for semantic search over internal documents.
- Shipped 3+ production RESTful APIs in FastAPI + SQLAlchemy as an intern, and was retained as a full-time engineer before graduation.
- Fixed N+1 query patterns and missing indexes in legacy SQL code, improving response times on key reporting endpoints.

TECHNICAL SKILLS

Languages: Python (AsyncIO), SQL
Agentic AI: Multi-Agent Orchestration, A2A Protocol, SPIFFE/SPIRE Identity, MCP Servers/Tools, Tool/Function Calling, Google ADK-style SDKs, Agent Memory, CopilotKit / AG-UI
LLM Evaluation & Safety: Golden Dataset Evals, LLM-as-a-Judge, Online/Offline Evaluation, Guardrails, Content Safety, Prompt-Injection Defense, GCP Cloud Armor, Langfuse Tracing
AI / LLM: LangChain, RAG Pipelines, Claude API, OpenAI API, sentence-transformers, pgvector, Prompt Engineering
Security: OAuth 2.0, OIDC, JWT, RBAC, Workload Identity Federation (WIF)
Backend: FastAPI, Django, SQLAlchemy, Celery
Databases: PostgreSQL, MySQL, SQL Server, MongoDB
Caching & Messaging: Redis, Kafka
Architecture: Microservices, REST APIs, Clean Architecture, System Design
Cloud & DevOps: Docker, GCP (Cloud Run, GKE), Kubernetes, Azure, CI/CD, Loki, Grafana

EDUCATION

B.E. Computer Science and Engineering 2020 – 2024
Meenakshi College of Engineering | CGPA: 8.06